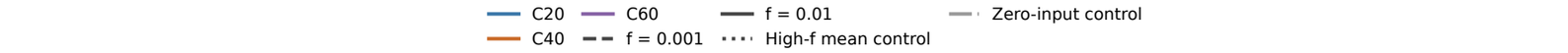
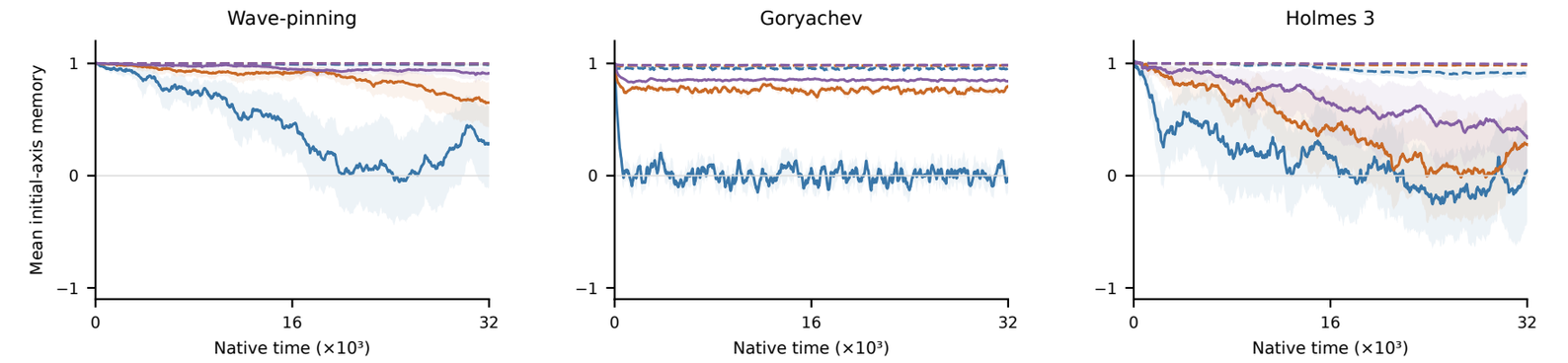
Figure 4 · Replicated activity-polarity memory under local fluctuations

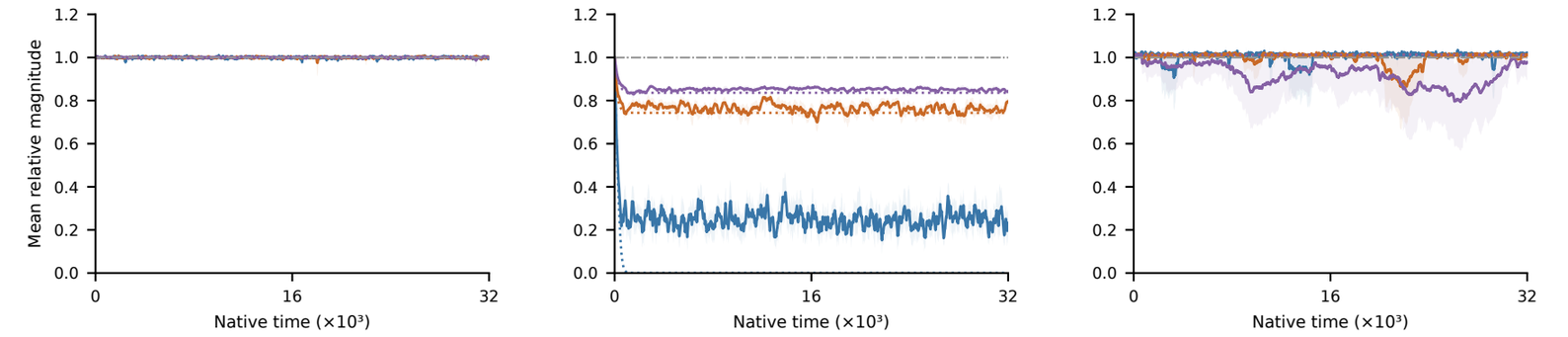
12 independent realizations per model/C/strength; C = 20, 40, 60; duration 32,000 native units. Added input is positive, with both declared means retained.
Fixed local statistics: μ / native cue peak = 0.001 or 0.01; Gaussian spatial parameter 1.5; normalized OU knots τ = 50, spacing 5. Native units differ by model.



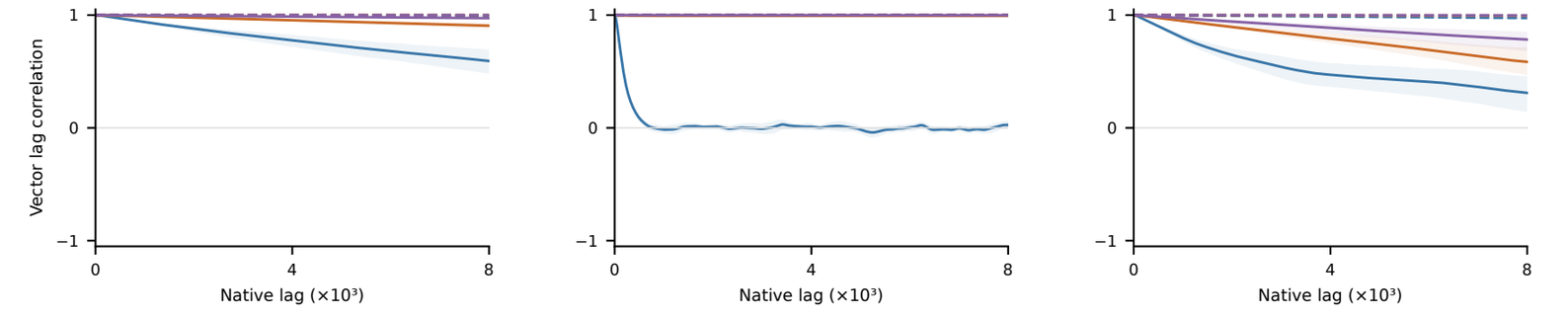
A Initial-axis memory ⟨Re P(t)⟩: both strengths; bands are 95% seed-bootstrap intervals



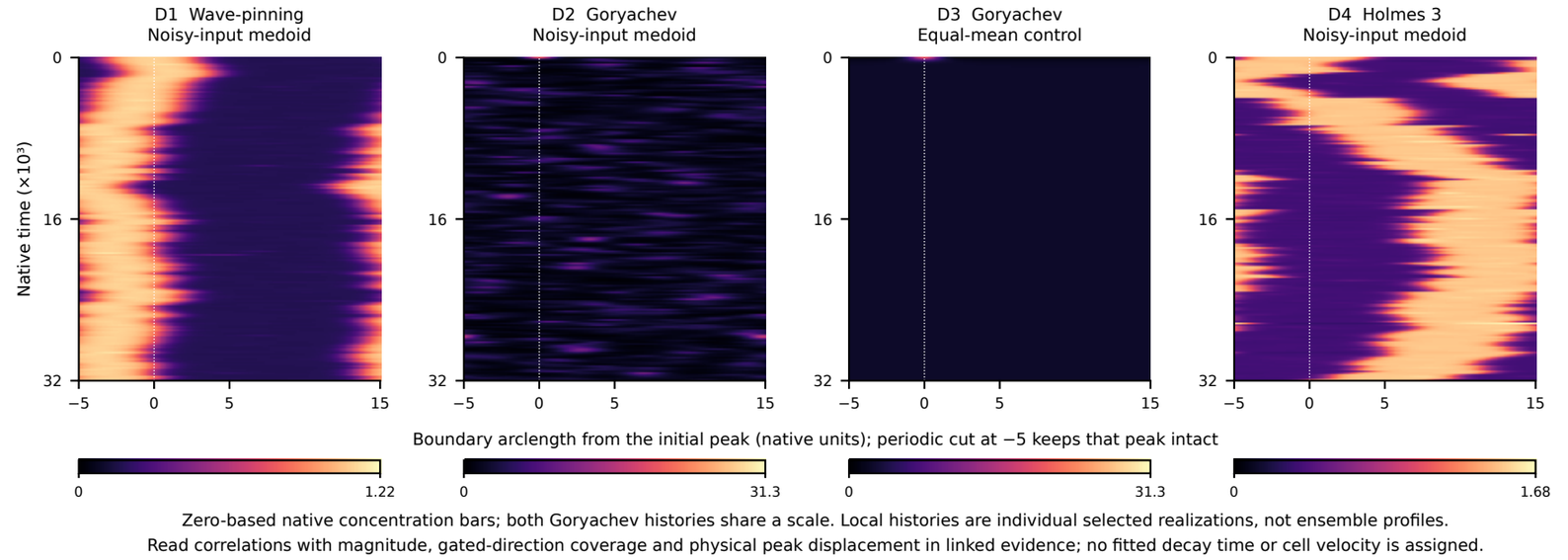
B High strength f = 0.01: mean magnitude ⟨|P|⟩ with deterministic-mean and zero-input controls



C Full-window vector lag correlation: both strengths; magnitude contributes to this unmasked observable



D Exact C20, f = 0.01 local histories: condition medoids and the equal-mean Goryachev control



Zero-based native concentration bars; both Goryachev histories share a scale. Local histories are individual selected realizations, not ensemble profiles.
Read correlations with magnitude, gated-direction coverage and physical peak displacement in linked evidence; no fitted decay time or cell velocity is assigned.